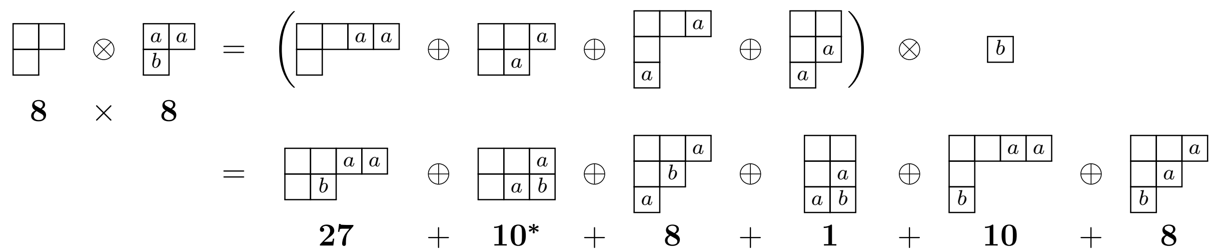
Homework 9: Solutions

Problem 1.

Evaluate the Clebsch-Gordan (CG) decomposition of the product of two adjoint matrices of $SU(3)$. Show that there are two adjoint matrices in this decomposition. Using that CG coefficients are invariant tensors, explain what the relation is between the two adjoint matrices in the decomposition, and the structure constants and the d-symbol. Could there be 3 adjoint matrices in the CG expansion of two adjoint matrices of $SU(N)$? One adjoint matrix? Show that for $SU(N)$ there are again always two adjoint matrices in the expansion.

Solution:

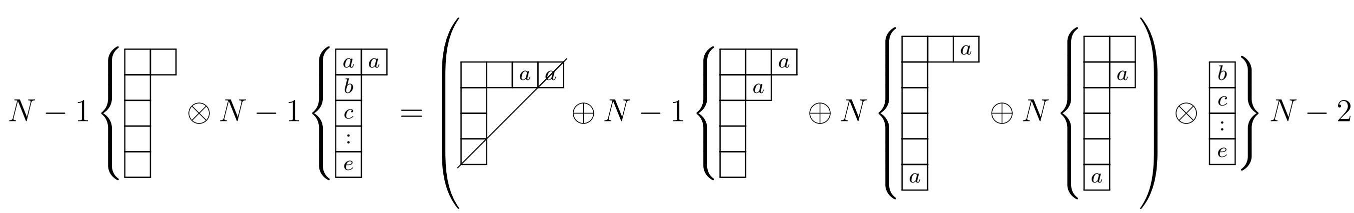


With tensors:

$$\begin{array}{cccccccccccc} u^\alpha_\beta v^\gamma_\delta & = & (uv)^{(\alpha\gamma)}_{(\beta\delta)} & + & (uv)^{(\alpha\gamma)}_{[\beta\delta]} & + & (uv)^{[\alpha\gamma]}_{(\beta\delta)} & + & (uv)^{[\alpha\gamma]}_{[\beta\delta]} & + & (u\cdot v)^\gamma_\beta & + & u:v \\ & & 6\times 6-9= & & 18-8= & & 18-8= & & \mathbf{8} & & \mathbf{8} & & \mathbf{1} \\ & & \mathbf{27} & & \mathbf{10} & & \mathbf{10^*} & & & & & & \end{array}$$

The term $(u\cdot v)^\alpha_\delta$ has the same structure as $(u\cdot v)^\beta_\gamma$, so we omitted it.

If one couples two octets, and if in the CG decomposition another octet appears, then the CG coupling constant for $8\times 8=8$ is an invariant tensor t_{abc}. There are two such tensors in $SU(N)$: the structure constants $f_{abc}\left(= f_{ab}{}^{c'} g_{c'c}\right)$ and the d-symbol d_{abc}. One can begin with the symmetric product $(8\times 8)_s$, and then $t_{abc}=d_{abc}$. On the other hand, $(8\times 8)_a$ yields f_{abc}. Since there are not 3 invariant tensors t_{abc}, one cannot obtain 3 adjoints in adjoint $\times$ adjoint. One can always project with f_{abc} or d_{abc} on adjoint $\times$ adjoint, so one should always find two adjoints in adjoint $\times$ adjoint for $SU(N)$. Direct evaluation of Young tableaux confirms this:



$$= \cdots \oplus \begin{ytableau} \none & & a \\ & a \\ & b \\ & c \\ & \\ e \end{ytableau} \oplus \begin{ytableau} & & a \\ & b \\ & c \\ & \\ & e \\ a \end{ytableau} \tag{1}$$

Using $\#a \geq \#b \geq \#c$, there are only two tableaux of the form

$$N \left\{ \ydiagram{3,2,2,2,2,1} \right. = N-1 \left\{ \ydiagram{2,1,1,1,1,1} \right. . \tag{2}$$

Problem 2.

Write down the CG decomposition of the product of two spinor irreps of $SO(10)$, and of the product of one spinor irrep and one conjugate spinor irrep. First construct the charge conjugation matrix in $d = 10$, and determine if it is block-diagonal or block-off-diagonal. Are the spinor and conjugate spinor irreps in $d = 10$ real, pseudoreal, or complex?

Solution: We begin by fixing the Dirac matrices in $d = 10 + 0$. We start from $d = 8 + 0$, and go via $d = 9 + 0$ to $d = 10 + 0$.

$\boldsymbol{\gamma}^{\mathbf{(8)}}$**:** 8 real symmetric 16×16 block-off-diagonal matrices. $\gamma_c^{(8)} = \gamma^1 \cdots \gamma^8$ satisfies $(\gamma_c^{(8)})^2 = I$, so $\gamma_c^8 = \gamma^9$ is also real and symmetric, but block-diagonal.

$\boldsymbol{\gamma}^{\mathbf{(9)}}$**:** $\{\gamma^{(8)}, \gamma_c^{(8)}\} = 9$ real symmetric 16×16 matrices.

$\boldsymbol{\gamma}^{\mathbf{(10)}}$**:** $\{\gamma^{(9)} \times \tau_1, I \times \tau_2\}$: 9 real symmetric matrices and one imaginary antisymmetric matrix, all in $d = 10 + 0$. All 10 matrices are hermitian and block-off-diagonal

$$\gamma^{(10)} = \left\{ \begin{pmatrix} 0 & \gamma^{(9)} \\ \gamma^{(9)} & 0 \end{pmatrix}, \begin{pmatrix} 0 & -iI \\ iI & 0 \end{pmatrix} \right\} \tag{3}$$

The charge conjugation matrices C_+ and C_- are as follows

$$\begin{aligned} C_+ \gamma^{(10)} &= \gamma^{10,T} C_+ = (\gamma^9 \times \tau_1, -I \times \tau_2) C_+ \Rightarrow C_+ = I \times \tau_1 \\ C_- \gamma^{(10)} &= -\gamma^{10,T} C_- = (-\gamma^9 \times \tau_1, I \times \tau_2) C_- \Rightarrow C_- = I \times (-i\tau_2) \end{aligned} \tag{4}$$

(One can also set $C_- = I \times \tau_2$, but $I \times (-i\tau_2)$ is real.) The $d = 10$ chirality matrix $\gamma_c^{(10)}$ is given by

$$\gamma_c^{(10)} \sim (\gamma^1 \times \tau_1)(\gamma^2 \times \tau_1) \cdots (\gamma^9 \times \tau_1)(I \times \tau_2) = I \times \tau_1 \tau_2 = I \times i\tau_3 \tag{5}$$

So $\gamma_c^{(10)} = I \times \tau_3$. Then $C_+ = C_- \gamma_c^{(10)}$ is a check.

$$C_+ = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}; \quad C_- = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \tag{6}$$

For the product of two spinors ψ^α and χ^α (with $\alpha = 1, \cdots, 32$, but $\chi^\alpha = \left(\begin{smallmatrix} \chi^a \\ 0 \end{smallmatrix}\right)$ and $\psi^\alpha = \left(\begin{smallmatrix} \psi^a \\ 0 \end{smallmatrix}\right)$ with $a = 1, \cdots, 16$) we need an odd number of Dirac matrices because $C_\pm$ and $\gamma^{(10)}$ are block-offdiagonal. This gives

$$\psi^\alpha \chi^\beta = \sum_{k=1,3,5} c_k (\gamma^{\mu_1 \cdots \mu_k} C^{-1})^{\alpha\beta} (\chi^T C \gamma_{\mu_1 \cdots \mu_k} \psi), \tag{7}$$

where c_k are constants. The spinor space of $\psi^\alpha \chi^\beta$ is $16 \times 16 = 256$ dimensional, and the Dirac matrices on the right-hand side yield a basis with $10 + \left(\frac{10 \cdot 9 \cdot 8}{1.2.3} = 120\right) + \left(\frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5} \frac{1}{2} = 126\right) = 256$ terms. The factor $\frac{1}{2}$ in the last term is due to the fact that an antisymmetric tensor $t^{\mu_1 \cdots \mu_5}$ of $SO(10)$ is reducible: it splits into two irreducible reps

$$\begin{aligned} t^{\mu_1 \cdots \mu_5} &= u^{\mu_1 \cdots \mu_5} + v^{\mu_1 \cdots \mu_5} \\ u^{\mu_1 \cdots \mu_5} &= \frac{1}{2} \left(t^{\mu_1 \cdots \mu_5} + \frac{i}{5!} \epsilon^{\mu_1 \cdots \mu_5 \mu_6 \cdots \mu_{10}} t_{\mu_6 \cdots \mu_{10}} \right) \\ v^{\mu_1 \cdots \mu_5} &= \frac{1}{2} \left(t^{\mu_1 \cdots \mu_5} - \frac{i}{5!} \epsilon^{\mu_1 \cdots \mu_5 \mu_6 \cdots \mu_{10}} t_{\mu_6 \cdots \mu_{10}} \right) \end{aligned} \tag{8}$$

For example

$$\begin{aligned} u^{12345} &= \frac{1}{2} \left(t^{12345} + i t^{678910} \right) \\ u^{678910} &= \frac{1}{2} \left(t^{678910} - i t^{12345} \right) = -i (u^{12345}) \end{aligned} \tag{9}$$

(The $-$ sign is due to $\epsilon^{\mu_1 \cdots \mu_5 \mu_6 \cdots \mu_{10}} = -\epsilon^{\mu_6 \cdots \mu_{10} \mu_1 \cdots \mu_5}$.) So there are only $\frac{256}{2}$ (complex) independent terms in u (and in v).

Next the CG decomposition of a chiral and an antichiral spinor (a spinor and a conjugate spinor) ψ^α and λ_α

$$\psi^\alpha \lambda_\beta = \sum_{k=0,2,4} c_l (\gamma^{\mu_1 \cdots \mu_l} C^{-1})^\alpha{}_\beta (\lambda^T C \gamma_{\mu_1 \cdots \mu_l} \psi), \tag{10}$$

The spinor space of $\psi^\alpha \lambda_\beta$ is again 256 dimensional, and the basis of Dirac matrices also has 256 terms

$$1 + \left(\frac{10 \cdot 9}{1.2} = 45 \right) + \left(\frac{10 \cdot 9 \cdot 8 \cdot 7}{1 \cdot 2 \cdot 3 \cdot 4} = 210 \right) = 256\,. \tag{11}$$

Finally we decide on the reality properties of the spinor irrep. The reality property of the spinor irrep is the same as that of the conjugate spinor irrep because they differ only by a sign for some of the generators, see (3)

$$\frac{1}{2}\gamma^{mn} = \frac{1}{2}\begin{pmatrix} \gamma^{ij}, i\gamma^{(9)} & 0 \\ 0 & \gamma^{ij}, -i\gamma^{(9)} \end{pmatrix} \tag{12}$$

with $i, j = 1, \cdots, 9$. If the irrep were real or pseudoreal we should have $(\gamma^{mn})^* = S\gamma^{mn}S^{-1}$. Hence $(i\gamma^j)^* = -i\gamma^j = Si\gamma^j S^{-1}$. Then $S\gamma^j + \gamma^j S = 0$, which would mean that in $d = 9+0$ there is a 10th DIrac matrix. This is not possible, hence the spinor (and conjugate spinor irrep) in $d = 10 + 0$ is **complex**.